

Traceable (Threshold) Secret Sharing: Detailed 3-Party Example (Using Shamir Secret Sharing)

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1. Scheme Setup

We work in the finite field $\mathbb{F}_7$ (elements $0, 1, \dots, 6$ with arithmetic mod 7). We set threshold $t = 3$ for $n = 3$ parties, so any three (honest) shares reconstruct the secret, while < 3 shares do not reveal the secret. We will trace a collusion of $f = t - 1 = 2$ parties.

Dealer's Polynomial: The dealer picks a random polynomial of degree $t - 1 = 2$ whose constant term is the secret $s = 3$:

$$q(X) = s + a_1X + a_2X^2 = 3 + 2X + X^2 \pmod{7}.$$

Share Generation: Choose distinct nonzero evaluation points $x_1 = 1, x_2 = 2, x_3 = 3$. Compute:

$$\begin{aligned} y_1 &= q(1) = 3 + 2 \cdot 1 + 1 \cdot 1^2 = 6, \\ y_2 &= q(2) = 3 + 2 \cdot 2 + 1 \cdot 2^2 = 11 \equiv 4, \\ y_3 &= q(3) = 3 + 2 \cdot 3 + 1 \cdot 3^2 = 18 \equiv 4. \end{aligned}$$

Each party P_i receives the share (x_i, y_i) together with a tracing key x_i .

2. Colluders' Reconstruction Box R

Parties P_1 and P_2 (indices 1 and 2) collude and build a black-box program R (reconstruction box) that hardcodes their shares $(1, 6)$ and $(2, 4)$. On input a candidate share (x^*, y^*) , R outputs the secret by Lagrange interpolation:

$$R(x^*, y^*) = \ell_1(x^*)6 + \ell_2(x^*)4 + \ell_3(x^*)y^*,$$

where

$$\ell_3(x^*) = \prod_{j=1}^2 \frac{x^* - x_j}{x_3 - x_j} = \frac{x^* - 1}{2} (x^* - 2) \pmod{7}.$$

3. Trace Procedure

The goal is to recover the tracing polynomial

$$h(X) = \prod_{k=1}^2 \frac{x_k - X}{x_k} = \frac{(1-X)(2-X)}{1 \cdot 2} = \frac{(X-1)(X-2)}{2} = \ell_3(X) \pmod{7},$$

whose roots $X = 1, 2$ identify the colluders. Since $\deg(h) = 2$, we need 3 evaluations $(x^*, h(x^*))$ to interpolate.

Because

$$\delta = R(x^*, y^* + 1) - R(x^*, y^*) = \ell_3(x^*),$$

we directly recover

$$h(x^*) = \delta.$$

3.1 Evaluation at Honest Share ($x^* = 3$)

Use the honest share $(3, 4)$:

- $s = R(3, 4) = 3$ (the true secret).
- $s' = R(3, 5) = R(3, 4 + 1) = 3 + \ell_3(3) \cdot 1$.
- Compute

$$\ell_3(3) = \frac{3-1}{2} (3-2) = 1 \cdot 1 = 1 \pmod{7}.$$

- Hence $\delta_1 = s' - s = 1$, giving

$$h(3) = 1.$$

3.2 Evaluation at Fresh Point ($x^* = 4$)

Pick arbitrary y^* (e.g. 0):

- $s = R(4, 0)$, $s' = R(4, 1) = s + \ell_3(4) \cdot 1$.
- Compute

$$\ell_3(4) = \frac{4-1}{2} (4-2) = \frac{3}{2} \cdot 2 = 3 \pmod{7}.$$

- Thus $\delta_2 = 3$, so

$$h(4) = 3.$$

3.3 Evaluation at Fresh Point ($x^* = 5$)

Similarly,

$$\ell_3(5) = \frac{5-1}{2} (5-2) = 2 \cdot 3 = 6, \quad h(5) = 6 \pmod{7}.$$

Collected Values:

$$(x^*, h(x^*)) = (3, 1), (4, 3), (5, 6).$$

4. Interpolation of the Tracing Polynomial

To obtain the polynomial coefficients, we substitute each evaluation point into

$$h(x) = ax^2 + bx + c$$

and set it equal to the value just computed. In particular, plugging $x = 3, 4, 5$ yields the system of linear congruences:

$$\begin{aligned} 9a + 3b + c &\equiv 1, \\ 16a + 4b + c &\equiv 3, \\ 25a + 5b + c &\equiv 6 \pmod{7}. \end{aligned}$$

Here each left-hand side is simply $ax^2 + bx + c$ evaluated at that x .

Reducing mod 7 gives:

$$\begin{aligned} 2a + 3b + c &= 1, \\ 2a + 4b + c &= 3, \\ 4a + 5b + c &= 6 \pmod{7}. \end{aligned}$$

Subtracting the first from the second yields $b = 2$. Subtracting the second from the third yields $2a + b = 3$, so $2a = 1$, hence $a = 4$. Finally,

$$2 \cdot 4 + 3 \cdot 2 + c = 1 \Rightarrow 8 + 6 + c = 1 \Rightarrow 14 + c \equiv c = 1 \pmod{7} \Rightarrow c = 1.$$

Therefore,

$$h(X) = 4X^2 + 2X + 1 \pmod{7},$$

whose roots $X = 1, 2$ identify the colluders. One checks that $h(1) = 7 \pmod{7} = 0$ and $h(2) = 21 \pmod{7} = 0$, recovering the colluders' identifiers. Also, $h(3) = 43 \pmod{7} = 1$, meaning the honest party also is not part of the corrupted indices.

5. Conclusion

By querying the reconstruction box R thrice (once at the honest share and twice at fresh points) and computing simple differences, we obtain $h(x^*)$ values, interpolate the degree-2 polynomial, and factor it to identify the traitors. Note that with large numbers however this becomes computationally harder to solve, since we heavily rely on the Chinese Remainder Theorem (CRT) as well as finding inverses in a group.